Abstract

Acknowledgements

Contents

List of Figures

List of Tables

Chapter 1

Strategic Vision and Project Scope

Contents

1.1 General Introduction

Digital transformation has changed how educational institutions manage teaching, communication, assessment, and academic follow-up. Despite this shift, many institutions still rely on fragmented combinations of messaging applications, file-sharing tools, video-conferencing platforms, and manually coordinated assessment processes. Such fragmentation reduces continuity in the learning experience, increases administrative overhead, and weakens the institutional ability to manage educational workflows within one controlled environment.

Eduverse is an integrated educational platform developed to address this problem through a unified, multi-organization, role-based environment. Its main implemented client is a web application, supported by an MVP mobile companion application that extends access to selected academic workflows for students and teachers. The platform is centered on the class as the primary digital workspace in which materials, assignments, examinations, results, notifications, live sessions, academic communication, and intelligent support services can be managed in a coordinated manner.

At a high level, Eduverse combines a modern web platform with supporting cloud and real-time services in order to provide a realistic and extensible educational system. The web client is built using Next.js, React, TypeScript, Tailwind CSS, and Supabase, while the mobile companion is built using Expo, React Native, and TypeScript. These technologies are mentioned here only as an introductory overview; detailed technical analysis and implementation discussion are reserved for later chapters.

1.2 Problem Statement

Educational institutions often manage closely related academic tasks through separate systems that are not designed to operate as one coherent platform. Materials may be distributed through one service, communication may take place in another, live teaching may depend on a third platform, and assessment records may be maintained elsewhere. As a result, students are forced to move between disconnected interfaces, while teachers and administrators struggle to maintain consistent oversight of participation, submissions, communication, and evaluation.

This problem becomes more serious in institutions that must manage multiple organizations, classes, and user roles under different access conditions. A practical educational platform must support not only learning activities, but also institutional control, membership management, and role-based access across different organizational contexts. Without these capabilities, digital education systems tend either to become rigid and limited or to become too loosely structured to support reliable academic administration.

A further challenge concerns platform coverage across devices. Rich authoring, administration, and class management workflows are best handled through a full web environment, whereas frequent day-to-day actions such as checking classes, following notifications, viewing assignments, opening materials, joining live sessions, and reading messages must also be accessible on mobile devices. This creates a need for an educational platform centered on a comprehensive web environment, supported by a practical mobile companion rather than a simplistic one-client-for-all approach.

1.3 Project Motivations

The development of Eduverse is motivated by the need to provide a unified academic environment in which essential educational workflows can be carried out without fragmentation. A coherent class-centered system can improve the student experience, reduce operational burden for teachers and administrators, and strengthen institutional visibility over communication, assessment, and academic progress.

The project is also motivated by the growing demand for flexible educational infrastructure. Modern institutions often require platforms that can support multiple organizations, distinct user roles, synchronous and asynchronous learning activities, and selective feature enablement within one extensible environment. Eduverse responds to this need by combining institutional control with practical support for daily teaching and learning workflows.

From an academic perspective, Eduverse is also motivated by its value as a software engineering project. It provides a realistic setting in which requirements analysis, interface planning, system integration, data management, real-time communication, AI-assisted services, and technical documentation can be studied and implemented within one coherent graduation project context.

1.4 Strategic Objectives

At the strategic level, Eduverse aims to establish an integrated educational platform whose primary implemented client is a web application, complemented by an MVP mobile companion application, in order to centralize essential academic workflows for organizations, teachers, and students within one role-aware and extensible system.

The main strategic objectives of the project are as follows:

- To centralize core educational workflows such as class management, materials, assignments, examinations, results, notifications, communication, and live sessions within one coordinated platform.
- To support multi-organization operation and role-based access for organization owners, organization admins, teachers, and students.
- To provide a primary web-based environment capable of supporting both institutional administration and day-to-day academic activity.
- To extend selected high-frequency academic workflows to a mobile MVP companion application without claiming full feature parity with the web platform.
- To incorporate distinctive educational capabilities such as an AI Agent and an integrated educational IDE as part of the platform's broader academic value.
- To organize data and services in a maintainable and scalable manner suitable for educational institutions with different operational needs.
- To produce a graduation project that is both academically meaningful and practically relevant as a modern educational software platform.

1.5 Educational and Professional Goals for the Team

Eduverse also serves clear educational and professional goals for the project team. From an educational standpoint, it provides an opportunity to apply software engineering concepts to a realistic problem domain rather than to an isolated classroom exercise. The team must analyze institutional needs, define system boundaries, organize requirements, structure user roles, and translate these decisions into a functional digital platform.

From a professional standpoint, the project strengthens practical experience in requirements analysis, frontend development, backend integration, database usage, real-time service integration, collaborative development workflows, and technical documentation. It therefore contributes not only to the production of software, but also to the development of disciplined engineering practice, teamwork, and academic reporting skills.

1.6 Project Scope and Boundaries

The scope of Eduverse in this dissertation is defined around the implemented web platform and the existing mobile MVP companion application. Chapter 1 is limited to introducing the problem context, strategic direction, main user groups, system boundaries, and general dissertation structure. Detailed requirements, architecture, database design, implementation decisions, and testing evidence are reserved for later chapters.

1.6.1 System Boundaries and User Roles

Eduverse operates within a multi-organization educational context and therefore distinguishes clearly between administrative, instructional, and learner responsibilities. At the institutional level, the platform supports organization owners and organization admins who manage organizations, user access, and operational control. At the instructional level, teachers manage class activities such as materials, assignments, live sessions, examinations, and academic follow-up. At the learner level, students access the classes to which they belong in order to study materials, follow announcements, submit work, join sessions, communicate, and review academic outcomes.

These role distinctions define important system boundaries. Eduverse is not a generic social platform in which all users have the same functional position. Instead, access is governed by organizational membership, class context, and academic responsibility. This role-aware structure supports accountability, controlled visibility, and more reliable institutional management.

1.6.2 Functional Scope

Within its current implemented scope, Eduverse includes authentication, multi-organization access, class management, materials, assignments, exams, results, notifications, live sessions, an AI Agent, and an integrated educational IDE. Together, these functions define the platform as more than a basic learning content portal. Eduverse is intended to support both academic management and active instructional workflows within one system.

At the same time, Chapter 1 remains introductory in nature. It does not attempt to document detailed database structures, API behavior, or low-level internal logic. Those aspects are part of the later analytical and technical chapters and lie outside the present scope.

1.6.3 Web and Mobile Client Scope

The main implemented Eduverse client is the web application. It provides the widest operational coverage of the platform, including organization-aware navigation, role-based access, class management, materials, assignments, exams, results, notifications, live sessions, AI Agent workflows, and the integrated educational IDE. For this reason, the web application should be regarded as the core working environment of the system.

The mobile application has a more focused scope. It is an MVP companion application intended to extend access to selected academic workflows such as authentication, dashboard viewing, classes, assignments, notifications, chat, materials, and live sessions. Although it is connected to the same broader platform environment, it is not yet feature-complete relative to the web application and should therefore be described as a companion application rather than a replacement for the primary platform.

1.6.4 Competitive Features and Technological Innovation

Eduverse is distinguished by the combination of features it brings together in one educational platform. A major distinguishing point is its multi-organization, role-based structure, which allows the same system to support organizational control, instructional management, and student participation in a coordinated way. The platform also integrates live sessions, an

AI Agent, and an educational IDE within the same broader academic environment rather than treating them as isolated external utilities.

Another important feature is the platform's attention to examinations and assessment integrity. Eduverse does not treat evaluation as a peripheral task, but as an integral part of the educational workflow that must be managed within a controlled digital environment. In addition, the system deliberately combines a comprehensive web platform with an MVP mobile companion application, thereby balancing rich institutional workflows with accessible everyday mobile use.

From a technological perspective, Eduverse also reflects a deliberate service separation strategy. Structured and frequently queried academic data are stored through Supabase PostgreSQL, large educational files are handled through AWS S3, and real-time live-session interaction is supported through LiveKit. This separated storage and service model strengthens the platform's practicality for an educational environment that must manage both relational records and larger real-time or file-based workloads.

1.7 Report Organization

This dissertation is organized as follows. Chapter 2 presents the literature review and theoretical framework. Chapter 3 describes the adopted methodology. Chapter 4 covers system analysis and requirements. Chapter 5 presents system design. Chapter 6 discusses software and engineering implementation. Chapter 7 presents testing and evaluation. Chapter 8 addresses future work and planned extensions, followed by the conclusion and appendices.

Chapter 2

Literature Review and Theoretical Framework

Contents

2.1 Introduction

This chapter presents the theoretical background that frames Eduverse as an integrated educational platform. It reviews the shift from traditional classroom management toward digital learning systems, highlights the limitations of fragmented educational tools, and situates Eduverse within the broader need for unified academic ecosystems. The chapter also discusses selected concepts from artificial intelligence in education, synchronous virtual learning, and secure role-aware platform design in order to establish the conceptual basis for the later analysis, design, and implementation chapters. It therefore draws selectively from the literature on Learning Management Systems, Artificial Intelligence in Education, Role-Based Access Control, and educational platform architecture.

2.2 Evolution of Digital Education Platforms

2.2.1 From Traditional Classroom Management to Digital Learning Systems

Educational management has historically depended on face-to-face coordination, manual attendance records, printed materials, and locally maintained assessment processes. As educational institutions expanded in scale and digital communication became more common, Learning Management Systems emerged to centralize course materials, announcements, assignment handling, and academic tracking within a unified digital environment [1, 2].

The move toward digital learning systems did not simply replace paper-based administration with electronic storage. It introduced new expectations about continuous access, asynchronous participation, online communication, and traceable academic workflows. Modern educational platforms are therefore judged not only by their capacity to store materials, but also by their ability to support coordination, interaction, assessment, and institutional oversight across different users and contexts [11].

2.2.2 Limitations of Fragmented Learning Management Tools

Although digital tools have become common in education, many institutions still depend on separate services for class management, communication, file sharing, video sessions, assessment, and student support. This fragmented model creates operational discontinuity because activities that belong to the same academic process are distributed across unrelated systems. As a result, students must move repeatedly between platforms, while teachers and administrators face a reduced ability to maintain clear academic oversight [11, 2].

Fragmentation also affects consistency and accountability. When communication, submissions, live sessions, and evaluation records are distributed across multiple tools, it becomes more difficult to maintain a coherent academic history, enforce clear access boundaries, and manage workflows in a structured way. From a system-design perspective, the limitation is not merely technical; it is also organizational, because fragmented tools weaken the integrity of the educational environment itself.

2.3 Research Gap and Proposed Integrated Learning Ecosystem

A clear gap exists between the broad range of activities required in modern education and the way those activities are often supported by disconnected systems. Institutions may rely on one tool for class coordination, another for messaging, another for live teaching, another for assignments, and still others for assessment, AI support, or programming practice. While each tool may be effective in isolation, the overall experience remains fragmented and operationally inefficient.

Eduverse addresses this gap by proposing an integrated learning ecosystem rather than a collection of separate utilities. Its core idea is that class management, communication, materials, assignments, live sessions, examinations, AI-assisted support, and programming practice should be coordinated within one platform governed by organizational structure and role-based access. In this sense, Eduverse is positioned as a response to fragmentation through integration, controlled scope, and educational workflow continuity.

This research direction is especially relevant because the platform does not attempt to treat every educational function as identical in weight or usage. Instead, it distinguishes between the main implemented web environment and the MVP mobile companion application, and between core academic workflows and specialized extensions such as AI support and the educational IDE. That balance between integration and scope control is central to the academic identity of the project.

2.4 Artificial Intelligence in Education

Artificial intelligence in education is commonly discussed as a means of augmenting teaching and learning rather than replacing teachers or institutional judgment [8, 6]. In educational settings, AI systems are valuable when they improve access to explanations, support learning continuity, reduce routine workload, and provide structured assistance without undermining academic responsibility. In Eduverse, this perspective frames the AI Agent as a bounded academic support service embedded in the platform rather than as an independent authority over teaching or evaluation.

2.4.1 Context-Aware Learning Assistance

One of the most important ideas in AI-supported education is that assistance is more useful when it is tied to the learner's actual academic context. Generic responses are often less effective than support that takes into account course materials, class discussion, ongoing tasks, and instructional expectations. Context-aware assistance can therefore improve relevance, reduce confusion, and support more focused academic interaction [6].

In relation to Eduverse, this concept is particularly important because the AI Agent is not conceived as an isolated chatbot detached from the broader learning environment. Its academic value depends on responding within the user's class, role, and ongoing learning activity so that assistance remains relevant to the immediate educational context. In practice, it is positioned as a support layer for understanding, clarification, and workflow assistance rather than as a substitute for teaching.

2.4.2 Material Summarization and Learning Support

Material summarization represents a practical form of AI assistance in education. Long documents, learning materials, and instructional resources can be difficult for students to process efficiently, especially when they are working under time constraints or trying to review several academic topics in parallel. AI-assisted summarization can support comprehension by highlighting main points, reducing information overload, and helping learners return more quickly to the central ideas of a topic [8].

This kind of support is most academically appropriate when it complements reading rather than replacing it. For that reason, summarization should be understood as a learning aid that improves navigation and revision rather than as an alternative to engagement with the full material. Within Eduverse, this type of assistance is most relevant for class materials, assignment instructions, and other course-related resources presented through the platform. The AI Agent may therefore support revision by summarizing or clarifying educational content while keeping the original material central.

2.4.3 Assessment and Feedback Support

AI can also support assessment-related workflows by assisting with draft guidance, formative feedback, and structured academic support around tasks and examinations. In theory, such support can help learners prepare, reflect on their work, and identify areas that require improvement [6]. However, educational literature also emphasizes that AI use in assessment must be governed carefully so that it supports learning without weakening academic integrity or teacher responsibility [8, 6].

For Eduverse, this means the AI Agent should assist students and teachers with preparatory guidance, clarification, and formative support around academic tasks while final instructional judgment and assessment responsibility remain with the teacher. This distinction is especially important in educational systems that must preserve trust, fairness, and instructor oversight.

2.5 Benchmarking: Fragmented Tools vs. Integrated Eduverse Approach

Table 2.1 summarizes the difference between a fragmented educational-tool model and the integrated approach proposed by Eduverse.

2.6 Theoretical and Technical Foundations of Eduverse

2.6.1 Web-Based Educational Platforms

Web-based educational platforms remain the strongest medium for full academic management because they support richer interfaces, broader administrative workflows, and deeper integration across organizational functions. They are especially suitable for tasks such as class configuration, materials management, assessment setup, and detailed instructional coordination.

Within Eduverse, this foundation helps justify the use of the web platform for workflows that require sustained academic coordination, detailed interaction, and institutional oversight.

2.6.2 Mobile Companion Learning Applications

Mobile educational applications play a different but still important role in digital learning environments. Their strength lies in quick access, timely participation, and support for routine high-frequency actions such as checking notifications, reviewing assignments, opening materials, and joining live sessions. As a result, they are especially useful as companion systems for daily academic engagement.

In Eduverse, the mobile client should therefore be understood as an MVP mobile companion rather than as a complete replacement for the main platform. Its purpose is to extend accessibility to core user-facing workflows while the more comprehensive management and authoring activities remain centered in the web application.

Table 2.1: Fragmented Educational Tools vs. Integrated Eduverse Approach

Educational Need	**Fragmented Tool Approach**	**Integrated Eduverse Approach**
Class management	Class records, memberships, and academic coordination are handled through separate administrative tools or partially manual processes.	Class organization is managed within one platform through organization-aware and role-based structures.
Communication	Messaging and announcements are distributed across external chat or notification tools.	Communication remains connected to the academic environment and class context.
Live sessions	Synchronous teaching depends on external meeting services disconnected from course workflows.	Live sessions are treated as part of the broader educational platform and linked to class participation.
Materials and assignments	Learning resources and tasks are often stored, shared, and submitted through different services.	Materials and assignments are coordinated within the same academic workspace.
AI support	AI tools are usually external and not directly tied to the learner's course context.	AI support is positioned within the class environment as contextual academic assistance.
Programming practice	Coding activities, when needed, often require separate platforms or disconnected lab tools.	Programming practice can be supported through an integrated educational IDE within the platform ecosystem.
Assessment	Examinations and grading may rely on detached tools with limited continuity or oversight.	Assessment workflows are integrated into the same role-aware educational system.

2.6.3 Real-Time Virtual Classrooms

Synchronous learning environments are an important part of modern educational theory because they strengthen immediacy, interaction, and instructional presence. When live classrooms are integrated into the broader academic platform rather than isolated in external services, continuity between teaching, communication, and follow-up becomes easier to maintain [5].

This foundation is relevant to Eduverse because live sessions are not treated as a separate product layer. Instead, they are part of the integrated learning environment and serve the broader goal of linking synchronous participation with the rest of the class workflow.

2.6.4 Integrated Coding Environments for Programming Education

Programming education often requires an environment in which learners can write code, inspect results, practice problem solving, and remain connected to the instructional context of the class. When coding practice is separated entirely from the educational platform, students may lose continuity between learning materials, teacher guidance, assignments, and practical execution.

An integrated coding environment addresses this problem by placing programming practice within the same educational ecosystem as the rest of the academic workflow. For Eduverse, the integrated educational IDE is therefore not merely a technical add-on, but part of the platform's broader pedagogical direction, especially for programming-related classes and technical learning settings.

2.6.5 Security, Roles, and Data Management

Trustworthy educational platforms depend on controlled access, clear role distinctions, and dependable data management. Role-Based Access Control is especially relevant in this context because it supports structured separation between administrative, teaching, and learner permissions [10, 4]. In educational systems, such distinctions are necessary not only for security, but also for accountability and appropriate institutional governance.

Data management is equally foundational. Educational systems must maintain reliable records, protect assessment-related information, and support different types of content ranging from structured academic data to larger learning materials and live-session interaction. From a theoretical perspective, this requires a design orientation that treats security, data organization, and role awareness as core educational infrastructure rather than as secondary technical concerns. This view is consistent with Eduverse, where access control, data separation, and institutional structure are part of the platform's academic trust model.

Taken together, the literature reviewed in this chapter supports Eduverse's design direction as an integrated educational platform with controlled scope, role-based access, AI-assisted learning support, live sessions, and programming-oriented practice within one academic environment. These foundations help explain why the project emphasizes coordinated educational workflows while preserving teaching responsibility, institutional control, and practical support for students and teachers.

Chapter 3

Methodology

Contents

3.1 Introduction

This chapter explains the methodology adopted for planning, analyzing, developing, and refining Eduverse as a graduation project. The emphasis is on the development process through which the project scope was transformed into a working educational platform, rather than on the detailed technical implementation of individual modules. The chapter therefore documents how the project moved from problem understanding and scope definition to system planning, feature development, service integration, and iterative refinement.

3.2 Adopted Development Methodology

Eduverse was developed using an iterative and incremental software development methodology. This approach was appropriate because the project combines several interdependent educational workflows, including organization management, class coordination, materials, assignments, examinations, notifications, live sessions, AI-assisted support, and programming practice. Rather than treating the system as a single linear implementation task, development progressed through repeated cycles of analysis, design, implementation, review, and adjustment.

The methodology was also feature-oriented. Core workflows were prioritized first, then expanded through successive refinements as the project structure became more stable. Within this process, the web application served as the main execution environment for the full platform, while the mobile application was developed as an MVP companion focused on selected high-frequency academic activities. This distinction supported scope control while preserving functional coherence between the two clients.

3.3 Justification for the Methodology

An iterative methodology was selected because Eduverse addresses a problem domain that is broad in function but sensitive to scope. The platform had to support multiple user roles, multiple organizations, synchronous and asynchronous learning activities, and several specialized services without turning the project into an unrealistic or weakly integrated system. A rigid linear process would have made it more difficult to validate assumptions about role behavior, workflow priorities, and the practical boundaries between the web platform and the mobile companion.

The chosen methodology also matched the academic nature of the project. Graduation projects typically benefit from gradual refinement because analysis, design, implementation, and documentation evolve together rather than as strictly isolated stages. For Eduverse, iterative development made it possible to stabilize the class-centered platform model early, then extend it carefully through live sessions, AI support, and the educational IDE while keeping the mobile scope controlled and realistic.

3.4 Development Phases

Although the project followed an iterative path, its progress can still be understood through a set of major development phases. These phases provide a structured view of how the project moved from conceptual problem analysis to a working integrated platform.

Table 3.1: Eduverse Development Phases

Phase	**Main Activities**	**Output**
Domain analysis	Studying fragmented educational workflows, user roles, and institutional needs	Initial problem model and platform direction
Scope definition	Identifying core features, priority users, and mobile MVP boundaries	Controlled project scope and feature priorities
Architecture planning	Organizing the system structure, data strategy, and external service roles	High-level architectural model
Web development	Building the primary academic workflows in the main client	Implemented web platform modules
Mobile development	Extending selected student and teacher workflows to the companion app	MVP mobile companion application
Integration and refinement	Connecting services, validating workflows, and correcting issues iteratively	More stable and coherent system behavior

3.4.1 Problem Understanding and Domain Analysis

The first phase focused on understanding the educational problem that Eduverse was intended to address. This included examining how materials, communication, assignments, live teaching, examinations, and academic follow-up are often distributed across separate tools. The analysis showed that fragmentation is not merely inconvenient; it weakens continuity, increases administrative overhead, and makes it harder for institutions to maintain a consistent view of academic activity.

This phase also clarified the domain structure of the platform. The class was identified as the central academic workspace, while the surrounding organizational context established the need for role-aware access and multi-organization support. The major user roles examined in this phase were organization owner, organization admin, teacher, and student, since these roles define both platform permissions and workflow expectations.

3.4.2 Requirement Collection and Scope Definition

After the problem domain was clarified, the next phase focused on translating that understanding into a controlled project scope. Functional expectations were centered on authentication, organization and class access, materials, assignments, examinations, results, notifications, live sessions, AI support, and programming-oriented learning support. At the same time, non-functional considerations such as maintainability, usability, responsiveness, and secure role separation were treated as necessary foundations rather than optional additions.

Scope definition was especially important because Eduverse includes both web and mobile clients. The main web platform was assigned responsibility for the full academic environment, including management and richer interactive workflows, whereas the mobile client was intentionally limited to MVP companion responsibilities such as authentication, dashboard access, classes, assignments, notifications, chat, materials, and live sessions. Advanced mobile parity was deliberately excluded from the current development scope.

3.4.3 System Design and Architecture Planning

Once the scope was defined, the project moved into high-level system planning. This phase established the platform as an integrated educational environment with clearly separated responsibilities across the client interface, secure server-side operations, structured data management, large-file handling, real-time communication, and AI-assisted support. The intention was not to over-engineer the system, but to create a structure capable of supporting multiple educational workflows without losing coherence.

Architecture planning also involved deciding how different types of platform data and services should be managed. Structured and frequently queried institutional data were aligned with Supabase PostgreSQL, large academic files were aligned with AWS S3, real-time live teaching sessions were aligned with LiveKit, and AI assistance was aligned with an external model-access layer. This planning phase created the conceptual foundation for the later implementation decisions described in subsequent chapters.

3.4.4 Web Application Development

The main body of development work was concentrated on the web application because it serves as the primary environment for the complete Eduverse platform. Development proceeded through feature-oriented iterations in which shared educational workflows were implemented, reviewed, and refined in relation to the user roles and class-centered structure identified earlier.

Within this phase, the web platform evolved to support the major academic functions of the system, including organization and role handling, class management, materials, assignments, examinations, results, notifications, live sessions, AI interaction, and the educational IDE. Although these features are discussed in more technical detail later, their methodological significance here lies in the fact that they were not treated as isolated pages. They were developed as connected parts of one integrated platform model.

3.4.5 Mobile Companion Development

The mobile application was developed as an MVP companion rather than as a full replication of the web platform. This methodological decision preserved the project's realism by recognizing that mobile interaction patterns differ from desktop-oriented academic management workflows. The mobile effort therefore focused on high-frequency participation tasks that are valuable in day-to-day educational use.

In practical terms, the mobile scope emphasized authentication, dashboard visibility, classes, assignments, notifications, chat, materials, and live session participation. This kept the companion application aligned with the main platform while avoiding unsupported claims of complete feature parity. Administrative workflows, advanced authoring, broader analytics, and richer management functions were intentionally left to the web platform or deferred as future work.

3.4.6 Backend, Database, and Service Integration

Another major phase concerned integrating the system components that allow the platform to operate as one coherent environment. This included secure authentication handling, API-based data exchange, role-aware access control, database-backed academic workflows, large-file access, live-session support, and AI service connectivity. Methodologically, this phase was important because integration quality directly affected whether Eduverse behaved like a unified platform or merely a group of adjacent features.

The project therefore adopted a service-aware integration strategy. Structured academic records were maintained in Supabase PostgreSQL, large learning files and submission-related assets were handled through AWS S3, real-time session capabilities were connected through LiveKit, and AI support was connected through OpenRouter. The mobile companion relied on the same identity and backend ecosystem, which helped preserve consistency between the two clients without duplicating platform ownership logic.

3.4.7 Testing and Refinement

The final recurring phase involved testing and refinement across the evolving system. Verification was not limited to one late-stage activity; it accompanied development through repeated checking of feature behavior, role restrictions, integration points, and interface consistency. This was especially important for workflows such as examinations, live sessions, and IDE-supported learning, where reliability affects both user trust and academic credibility.

The refinement process combined targeted automated checks with iterative manual review. Available tests and quality checks focused on selected core logic such as examination behavior, IDE runners, feature selection logic, and related supporting components, while manual review remained important for integrated user flows across roles and devices. Full end-to-end automation was outside the current project scope, so iterative correction and controlled verification were used to stabilize the implemented platform.

3.5 Tools and Technologies Used in the Development Process

The development process relied on a technology set selected to support a modern, maintainable educational platform without fragmenting the project into unrelated systems. The main implemented client was built as a web application using Next.js, React, TypeScript, and Tailwind CSS. The companion mobile application was built using Expo, React Native, and TypeScript in order to support selected mobile workflows within the same broader platform direction.

For data and service integration, the project used Supabase for authentication and structured platform data, AWS S3 for larger academic files, LiveKit for real-time live sessions, and OpenRouter for AI Agent connectivity. These technologies are identified here as part of the methodological environment of the project; their detailed technical roles are discussed in later chapters.

3.6 Team Workflow and Project Management

The project workflow emphasized continuous coordination between analysis, implementation, and documentation. Rather than separating the work into fully independent tracks, the team progressed through repeated cycles in which requirements were clarified, features were prioritized, implementation choices were reviewed against scope, and documentation was updated to reflect verified project realities.

Priority was consistently given to the workflows that define Eduverse as an integrated educational platform. This meant that class-centered activities, role-sensitive access, assessment support, live sessions, and controlled mobile extension received more attention than peripheral or purely decorative features. Such prioritization helped the team preserve academic relevance and avoid uncontrolled expansion beyond the graduation-project scope.

3.7 Methodological Challenges

One of the main methodological challenges in Eduverse was balancing feature breadth with controlled scope. The platform combines educational management, communication, assessment, real-time teaching, AI support, and programming practice, all of which could easily lead to over-expansion. This challenge was addressed by keeping the web platform as the main implementation target and by treating the mobile client as an MVP companion with intentionally limited responsibilities.

Another challenge concerned integration complexity. Multi-organization access, role-aware behavior, live sessions, storage separation, and AI support all required careful coordination so that the system would remain coherent rather than fragmented internally. A further challenge was maintaining accurate documentation boundaries, especially when distinguishing between features that are already implemented and features that remain under development. These constraints reinforced the importance of iterative refinement and disciplined scope management throughout the project.

3.8 Summary

This chapter has described the methodology through which Eduverse was planned, developed, integrated, and refined as a graduation project. The project followed an iterative, feature-oriented approach that supported gradual validation of scope, roles, services, and user workflows. This methodological foundation prepares the transition to the next chapter, which examines the system analysis and requirement structure in greater detail.

Chapter 4

System Analysis and Requirements

TODO: This chapter outline is reserved for a later documentation step.

4.1 Introduction

4.2 Problem Domain Analysis

4.3 Stakeholder Needs Analysis

4.4 As-Is Analysis

4.5 To-Be Analysis

4.6 Requirements Engineering

4.6.1 Functional Requirements

4.6.2 Non-Functional Requirements

4.7 Use Case Modeling

4.7.1 Use Case Actors

4.7.2 Use Case Diagrams

4.7.3 Planned Extension Use Cases

4.8 Activity Diagrams

4.8.1 Access and Onboarding

4.8.2 Organization and Configuration Management

4.8.3 Classroom Operations

4.8.4 Assessment Workflows

4.8.5 Synchronous Learning and Follow-Up

Chapter 5

System Design

TODO: This chapter outline is reserved for a later documentation step.

5.1 Introduction

5.2 Architectural Design Overview

5.3 Data and Storage Design

5.4 Interface and Interaction Design

5.5 Integration and Security Design

Chapter 6

Software and Engineering Implementation

TODO: This chapter outline is reserved for a later documentation step.

6.1 Introduction

6.2 Web Application Implementation

6.3 Mobile Companion Implementation

6.4 Backend and Service Integration

6.5 Engineering Challenges and Resolutions

Chapter 7

System Testing and Evaluation

TODO: This chapter outline is reserved for a later documentation step.

7.1 Introduction

7.2 Testing Strategy

7.3 Functional Testing

7.4 Non-Functional Evaluation

7.5 Discussion of Results

Chapter 8

Future Work and Planned Features

TODO: This chapter outline is reserved for a later documentation step.

8.1 Introduction

8.2 Mobile Expansion

8.3 AI Agent and IDE Enhancement

8.4 Scalability and Institutional Growth

Chapter 9

Conclusion

TODO: Write the final conclusion after the main report chapters are completed.

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Appendix A

Appendices